Weak forms and problem setup — 2D plate with a circular hole under tension

1 Governing equations

1.1 Geometry and boundary

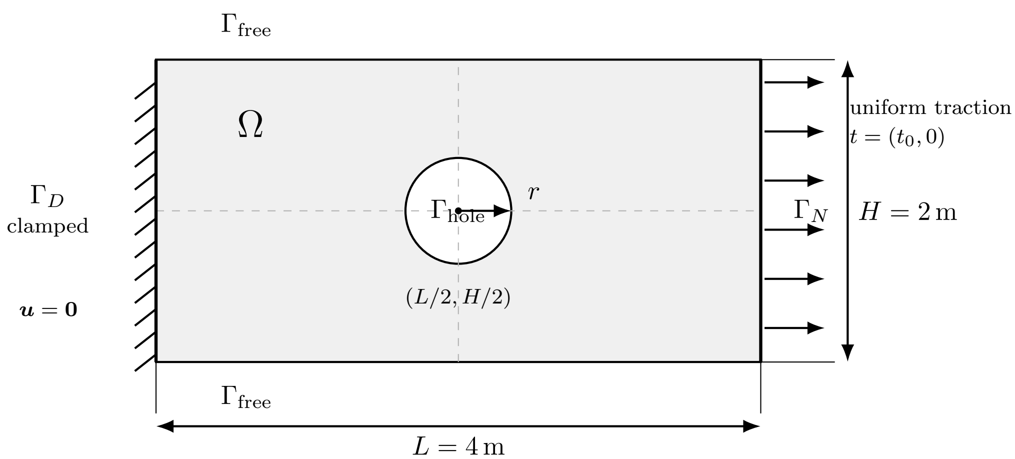

Figure 1: Plate domain Ω with a circular hole of radius $r = 0.35\,\text{m}$ centered at $(L/2, H/2)$. The left edge Γ_D is clamped, the right edge Γ_N is subjected to the uniform traction $t = (t_0, 0)$, and the top, bottom, and hole boundaries are traction-free.

Let $\Omega \subset \mathbb{R}^2$ denote the perforated rectangular plate

$$\Omega := \left\{ (x, y) \in (0, L) \times (0, H) \;\middle|\; \left(x - \frac{L}{2}\right)^2 + \left(y - \frac{H}{2}\right)^2 > r^2 \right\},$$

with

$$L = 4\,\text{m}, \qquad H = 2\,\text{m}, \qquad r = 0.35\,\text{m}.$$

The boundary is decomposed as

$$\begin{aligned}
\Gamma_D &:= \{(x, y) \in \partial\Omega : x = 0\}, \\
\Gamma_N &:= \{(x, y) \in \partial\Omega : x = L\}, \\
\Gamma_{\text{hole}} &:= \left\{ (x, y) \in \partial\Omega : \left(x - \frac{L}{2}\right)^2 + \left(y - \frac{H}{2}\right)^2 = r^2 \right\}, \\
\Gamma_{\text{free}} &:= \partial\Omega \setminus (\Gamma_D \cup \Gamma_N \cup \Gamma_{\text{hole}}).
\end{aligned}$$

1.2 Material parameterization

The simulation parameters are Young's modulus E and Poisson ratio ν, sampled over the ranges

$$E \in [2.0, 10.0], \qquad \nu \in [0.20, 0.45].$$

These values are interpreted in the consistent unit system used throughout the example.
The corresponding 3D Lamé parameters are

$$\lambda = \frac{E\nu}{(1+\nu)(1-2\nu)}, \qquad \mu = \frac{E}{2(1+\nu)}.$$

Under the plane-stress condition $\sigma_{zz} = 0$, the effective in-plane constitutive law uses

$$\bar{\lambda} = \frac{2\mu\lambda}{\lambda + 2\mu} = \frac{E\nu}{1-\nu^2}.$$

Hence, for the in-plane displacement field $\boldsymbol{u} : \Omega \to \mathbb{R}^2$, the Cauchy stress is

$$\boldsymbol{\sigma}(\boldsymbol{u}) = 2\mu\, \boldsymbol{\varepsilon}(\boldsymbol{u}) + \bar{\lambda}\, \mathrm{tr}(\boldsymbol{\varepsilon}(\boldsymbol{u}))\, \mathbf{I},$$

where $\mathbf{I}$ is the 2×2 identity tensor.

1.3 Governing laws

The small-strain tensor is

$$\boldsymbol{\varepsilon}(\boldsymbol{u}) := \frac{1}{2}\left(\nabla \boldsymbol{u} + \nabla \boldsymbol{u}^\top\right) \in \mathbb{R}^{2\times 2}.$$

The traction vector on a boundary with outward unit normal $\boldsymbol{n}$ is

$$\boldsymbol{t} := \boldsymbol{\sigma}(\boldsymbol{u})\boldsymbol{n}.$$

In the absence of body forces, static equilibrium requires

$$-\nabla \cdot \boldsymbol{\sigma}(\boldsymbol{u}) = \mathbf{0} \qquad \text{in } \Omega.$$

If a consistent SI interpretation is adopted, then

$$[\boldsymbol{\sigma}] = \mathrm{N/m}^2, \qquad [\nabla \cdot \boldsymbol{\sigma}] = \mathrm{N/m}^3.$$

1.4 Governing PDE

Let

$$V_{\text{strong}} := [C^0(\overline{\Omega}) \cap C^2(\Omega)]^2.$$

The strong form reads: find $\boldsymbol{u} \in V_{\text{strong}}$ such that

$$-\nabla \cdot \boldsymbol{\sigma}(\boldsymbol{u}) = \mathbf{0} \qquad \text{in } \Omega, \tag{1}$$
$$\boldsymbol{u} = \mathbf{0} \qquad \text{on } \Gamma_D, \tag{2}$$
$$\boldsymbol{\sigma}(\boldsymbol{u})\boldsymbol{n} = (t_0, 0), \qquad t_0 = 1.0 \qquad \text{on } \Gamma_N, \tag{3}$$
$$\boldsymbol{\sigma}(\boldsymbol{u})\boldsymbol{n} = \mathbf{0} \qquad \text{on } \Gamma_{\text{hole}} \cup \Gamma_{\text{free}}. \tag{4}$$

Here,

$$\boldsymbol{\sigma}(\boldsymbol{u}) = 2\mu\, \boldsymbol{\varepsilon}(\boldsymbol{u}) + \bar{\lambda}\, \mathrm{tr}(\boldsymbol{\varepsilon}(\boldsymbol{u}))\, \mathbf{I}, \qquad \boldsymbol{\varepsilon}(\boldsymbol{u}) = \frac{1}{2}\left(\nabla \boldsymbol{u} + \nabla \boldsymbol{u}^\top\right).$$

2 Weak form

2.1 Variational statement

Define the test and trial space

$$V := \left\{ \boldsymbol{v} \in [H^1(\Omega)]^2 \;\middle|\; \boldsymbol{v} = \mathbf{0} \text{ on } \Gamma_D \right\}.$$

Multiplying the equilibrium equation by $\boldsymbol{v} \in V$, integrating over Ω, and applying integration by parts yields

$$\int_\Omega \boldsymbol{\sigma}(\boldsymbol{u}) : \nabla \boldsymbol{v} \; \mathrm{d}\mathbf{x} - \int_{\partial\Omega} (\boldsymbol{\sigma}(\boldsymbol{u})\boldsymbol{n}) \cdot \boldsymbol{v} \; \mathrm{d}s = 0.$$

Using the symmetry identity

$$\boldsymbol{\sigma}(\boldsymbol{u}) : \nabla \boldsymbol{v} = \boldsymbol{\sigma}(\boldsymbol{u}) : \boldsymbol{\varepsilon}(\boldsymbol{v}),$$

together with the boundary conditions and the fact that $\boldsymbol{v} = \mathbf{0}$ on Γ_D, the weak problem becomes: find $\boldsymbol{u} \in V$ such that

$$a(\boldsymbol{u}, \boldsymbol{v}; E, \nu) = \ell(\boldsymbol{v}) \qquad \forall\, \boldsymbol{v} \in V,$$

where

$$a(\boldsymbol{u}, \boldsymbol{v}; E, \nu) := \int_\Omega \Big[2\mu\, \boldsymbol{\varepsilon}(\boldsymbol{u}) : \boldsymbol{\varepsilon}(\boldsymbol{v}) + \bar{\lambda}\; \mathrm{tr}(\boldsymbol{\varepsilon}(\boldsymbol{u}))\; \mathrm{tr}(\boldsymbol{\varepsilon}(\boldsymbol{v})) \Big] \; \mathrm{d}\mathbf{x}, \tag{5}$$

and

$$\ell(\boldsymbol{v}) := \int_{\Gamma_N} \boldsymbol{t} \cdot \boldsymbol{v} \; \mathrm{d}s = \int_{\Gamma_N} t_0\, v_x \; \mathrm{d}s. \tag{6}$$

2.2 Affine decomposition

Because the domain consists of a single uniform material region, stored in the implementation as `region_1`, the stiffness form admits the two-term affine decomposition

$$a(\boldsymbol{u}, \boldsymbol{v}; E, \nu) = \theta_1(E, \nu)\, a_1(\boldsymbol{u}, \boldsymbol{v}) + \theta_2(E, \nu)\, a_2(\boldsymbol{u}, \boldsymbol{v}), \tag{7}$$

with

$$\theta_1(E, \nu) = \bar{\lambda}(E, \nu) = \frac{E\nu}{1 - \nu^2}, \qquad \theta_2(E, \nu) = \mu(E, \nu) = \frac{E}{2(1 + \nu)},$$

and

$$a_1(\boldsymbol{u}, \boldsymbol{v}) = \int_\Omega \mathrm{tr}(\boldsymbol{\varepsilon}(\boldsymbol{u}))\; \mathrm{tr}(\boldsymbol{\varepsilon}(\boldsymbol{v}))\; \mathrm{d}\mathbf{x},$$

$$a_2(\boldsymbol{u}, \boldsymbol{v}) = \int_\Omega 2\, \boldsymbol{\varepsilon}(\boldsymbol{u}) : \boldsymbol{\varepsilon}(\boldsymbol{v}) \; \mathrm{d}\mathbf{x}.$$

The linear functional is parameter-independent because the traction magnitude t_0 is fixed. Accordingly, the full-order stiffness matrix can be written as

$$\mathbf{K}(E, \nu) = \bar{\lambda}(E, \nu)\, \mathbf{K}_1 + \mu(E, \nu)\, \mathbf{K}_2. \tag{8}$$

Block	Coefficient	Parameter-independent contribution
$\mathbf{K}_1$	$\bar{\lambda}(E, \nu) = \dfrac{E\nu}{1 - \nu^2}$	$\int_\Omega \mathrm{tr}(\boldsymbol{\varepsilon}(\boldsymbol{u}))\; \mathrm{tr}(\boldsymbol{\varepsilon}(\boldsymbol{v}))\; \mathrm{d}\mathbf{x}$
$\mathbf{K}_2$	$\mu(E, \nu) = \dfrac{E}{2(1 + \nu)}$	$\int_\Omega 2\, \boldsymbol{\varepsilon}(\boldsymbol{u}) : \boldsymbol{\varepsilon}(\boldsymbol{v}) \; \mathrm{d}\mathbf{x}$

This is the affine structure used for the offline/online reduced-order decomposition.

3 Finite element discretization and ROM

3.1 Finite element discretization used in the code

The implementation discretizes the plate on a triangular mesh generated from a structured point cloud followed by Delaunay triangulation. Triangles whose centroids lie inside the circular hole are removed, unused vertices are pruned, and the mesh is then uniformly refined.
The code therefore uses

- a 2D triangular mesh (`MeshTri`),
- first-order scalar triangular basis functions (`ElementTriP1`),
- a vector-valued displacement space through `ElementVector(ElementTriP1())`, and
- numerical boundary identification for the hole using a radial tolerance.

Accordingly, the circular hole in the computation is represented by a polygonal approximation of the exact circular boundary shown in the schematic.